Mississippi Valley State University

# CS 341 — Discrete Structures

## Lecture 23: When Direct Proof Stalls

*Proof by Contrapositive*

Chartrand & Zhang, Section 3.3

Name: _____ Date: _____

**Where we are**

Wednesday's method works when the hypothesis gives you something to compute with. Sometimes it does not — and then the same statement, written backwards and negated, becomes easy. You have had the tool since Week 2.

### By the end of today you should be able to

1. Recognise when a direct proof will stall.
2. Write a proof by contrapositive with the correct opening.
3. Spot a proof that proves the wrong implication.

### 1. The method

**Proof by contrapositive**

To prove $P \Rightarrow Q$, instead prove

$$(\sim Q) \Rightarrow (\sim P)$$

by a *direct* proof: assume $\sim Q$, derive $\sim P$.

This is legitimate because of _____ — an implication and its contrapositive are logically equivalent, so proving either proves the other.

You verified that with a truth table in Week 2 and used it as a shortcut in Week 7. Today it becomes a proof technique in its own right.

### 2. Why you need it

**Claim**

Let $n$ be an integer. If $n^2$ is even, then $n$ is even.

**Try direct.** Assume $n^2$ is even, so $n^2 = 2k$ for some integer $k$. Then $n = \sqrt{2k}$ — and now what? You need $n$ to be an integer of the form $2a$, and a square root gives you _____. The direct route

stalls.

**Try contrapositive.** The contrapositive is: if $n$ is _____, then $n^2$ is _____.

*Proof.* Assume $n$ is odd, so $n = 2b + 1$ for some integer $b$. Then

$$n^2 = (2b+1)^2 = 4b^2 + 4b + 1 = 2(_____) + 1.$$

Since $2b^2 + 2b$ is an integer, $n^2$ is odd. ■

Three lines, and every step is the Wednesday template. **The hypothesis was the problem, not the mathematics** — "$n$ is odd" hands you an equation, and "$n^2$ is even" does not.

**Theorem 3.24**

Combining this with the direct proof of the converse gives:

Let $n \in \mathbb{Z}$. Then $n^2$ is even if and only if $n$ is even.

A biconditional needs _____ proofs, and here one goes direct and one goes contrapositive. **The two halves need not use the same method.**

## 3. When to reach for it

**The signal**

Reach for the contrapositive when the **hypothesis is hard to use** and the **conclusion is easy to negate**.
Typical triggers: the hypothesis involves $n^2$, $n^3$, or a product, while the conclusion is a simple parity or sign statement. Negating swaps them, and the easy one becomes your starting point.

**Your turn.** Prove: if $3n + 11$ is odd, then $n$ is even.

*Contrapositive:* if $n$ is _____, then $3n + 11$ is _____.

*Proof.* Assume $n$ is odd, so $n = 2k + 1$. Then

$$3n + 11 = 3(2k+1) + 11 = 6k + 14 = 2(_____).$$

Since $3k + 7$ is an integer, $3n + 11$ is even. ■

## 4. A proof that proves the wrong thing

Consider this claim and its offered proof.

**Claim**

Let $n \in \mathbb{Z}$. Then $5n + 7$ is even only if $n$ is odd.

**Offered proof — what is wrong with it?**

*Assume first that $n$ is odd. Then $n = 2a + 1$, so $5n + 7 = 10a + 12 = 2(5a + 6)$, hence $5n + 7$ is even.*
*Next assume $n$ is even. Then $n = 2b$, so $5n + 7 = 10b + 7 = 2(5b + 3) + 1$, hence $5n + 7$ is odd.*

**What went wrong.** Recall from Week 2 that "$P$ only if $Q$" means _____. So the claim is: *if $5n+7$ is even, then $n$ is odd.*

The first paragraph proves the _____. The second proves the _____ of the converse — which is the converse again.

**Neither paragraph proves the stated claim.** The algebra is flawless and the proof is still wrong.

**The lesson**

Before writing anything, **state precisely which implication you are proving** and what your opening assumption therefore is. Correct algebra attached to the wrong implication earns nothing — and it is much harder to spot than an arithmetic slip.

## Exit ticket

1. Write the contrapositive of: "if $7n-3$ is even, then $n$ is odd."

2. In a proof by contrapositive of $P \Rightarrow Q$, what do you assume and what do you derive?

## Before Monday

**Review** Chapters 1, 2, and 5 for the midterm. Monday, October 12 is a review session — bring questions.

**Self-check (not collected).** Section 3.3, Exercises 1, 3, 5, and 9 — odd-numbered, answers in the back. Exercise 14 is the flawed proof we discussed.

**Coming up**

**Homework 8** is due tonight, Friday, October 9, 11:59 p.m.
**Midterm: Wednesday, October 14** — Chapters 1, 2, and 5. **No quiz Monday**; the review session replaces it.
**Homework 9** covers Sections 5.5, 3.2, and 3.3, due Friday, October 23.

## Questions I still have